

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS Computer Science candidate (NC State, May 2026) with hands-on experience in ranking systems, retrieval pipelines, and LLM evaluation. Built VectorLift, a production-grade semantic search engine on MS MARCO with BM25, fine-tuned bi-encoder dense retrieval, and cross-encoder re-ranking -- evaluated across 6 pipeline configurations with paired bootstrap significance testing. Shipped multi-model AI pipelines across 7 production deployments at Isomer AI (394 commits) and built batch data pipelines with Apache Spark and Kafka. Practical ML engineer: no research, just systems that ship.

CORE COMPETENCIES

Ranking + Retrieval Systems

PyTorch + Fine-Tuning

LLM Evaluation + Significance Testing

Apache Spark + Kafka Pipelines

Elasticsearch + Vector Search

RAG + Dense Retrieval

FastAPI + Streamlit

Docker + Prometheus + CI/CD

WORK EXPERIENCE

Isomer AI

May 2025 - Present

Product Engineer Intern | Remote

- Built and maintained **multi-model AI pipelines** across 8+ models (Claude, GPT-4o, Gemini) with MCP tooling, Temporal.io workflow orchestration, and real-time event streaming -- 394 commits across 7 production deployments.
- Implemented document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) and role-based access control across 10+ shipped backend features for an AI-powered claims automation platform.

NC State University

April 2025 - Present

Graduate Research Assistant | Raleigh, NC

- Designed a cloud-based genomic data platform (JBrowse 2) for interactive exploration of 500GB datasets, serving 100+ researchers on Google Cloud Run.
- Collaborated on ML architecture design with hyperparameter tuning and time-series analysis, improving model accuracy by 12%.

Mphasis (HP Company)

Jun 2024 - Aug 2024

Software Developer Intern | Remote

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML classification models -- improved processing accuracy by 25%.
- Automated deployments on AWS and GCP with Grafana monitoring; supported model validation and red-teaming to improve ML reliability.

Capgemini

Jun 2023 - Aug 2023

AI/ML Intern | Remote

- Built end-to-end **data pipelines** using Apache Hadoop and Spark for batch processing and Kafka for real-time streaming -- reduced ticket-processing latency by 20%.
- Led SDLC for a support ticket NLP classification system, reducing manual processing time by 15%.

PROJECTS

VectorLift -- Semantic Search + Ranking Engine MS MARCO / PyTorch

Built a production-grade search system on MS MARCO: BM25 baseline (Elasticsearch), dense bi-encoder retrieval (sentence-transformers fine-tuned with in-batch negatives), and cross-encoder re-ranker (MiniLM). Evaluated 6 pipeline configurations -- BM25, dense, hybrid, BM25+rerank, dense+rerank, hybrid+rerank -- with NDCG@10, MRR@10, MAP, and paired bootstrap significance testing to identify statistically significant wins. Kafka-driven indexing pipeline feeds Elasticsearch and Qdrant; FastAPI serves ranked results with per-stage latency; Streamlit dashboard surfaces experiment comparisons and significance outputs.

Python · PyTorch · Sentence-Transformers · Elasticsearch · FAISS · Qdrant · Kafka · FastAPI · Streamlit · PostgreSQL · Redis · Prometheus · Grafana · Docker Compose | github.com/Aditya-k24/VectorLift

LLM Eval Harness Statistical ML Evaluation

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types (grounded QA, multi-hop QA, FEVER claim verification) on a 2,000-example frozen test set. Tracked ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call. Statistical significance via paired bootstrap CIs, McNemar test, and Holm correction.

Python · Anthropic/OpenAI/Google SDKs · DuckDB · Parquet · Streamlit · Docker | github.com/Aditya-k24/llm-eval-harness

AlgoPulse -- Agentic AI Orchestration Platform Temporal + Kafka

Event-driven AI orchestration platform scaled to 2,000 async inference requests/second. Identified API-polling bottleneck under load and pivoted to Supabase Edge Functions streaming with Temporal.io for workflow state management and Kafka for fault-tolerant synchronization -- zero state loss under load.

TypeScript · Node.js · React · Supabase Edge Functions · Temporal.io · Kafka | github.com/Aditya-k24/AlgoPulse

Glance AI -- Privacy-First RAG System RAG / Vector Retrieval

Compliance-ready RAG retrieval system for fintech and healthcare with per-tenant data isolation at the vector retrieval layer, preventing cross-tenant context leakage and LLM hallucinations. Deterministic output architecture for regulated environments.

Python · LLMs · Vector Databases · RAG | github.com/Aditya-k24/GlanceAI

EDUCATION

MS Computer Science -- NC State University Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms

BTech Computer Science -- Mumbai University Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Warehousing, Big Data Infrastructure, Machine Learning, DS & Algorithms

SKILLS

ML/AI: Python, PyTorch, Sentence-Transformers, Scikit-learn, Hugging Face Transformers, LangChain, RAG, Pandas, NumPy, OpenAI API, MCP

Search/Retrieval: Elasticsearch, FAISS, Qdrant, BM25, dense retrieval, re-ranking, hybrid search

Data Pipelines: Apache Spark, Hadoop, Kafka, DuckDB, Parquet, Prefect

Backend: FastAPI, TypeScript, Java, Node.js, NestJS, Spring Boot, PostgreSQL, Redis, MySQL, MongoDB

Cloud/DevOps: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Prometheus, Grafana